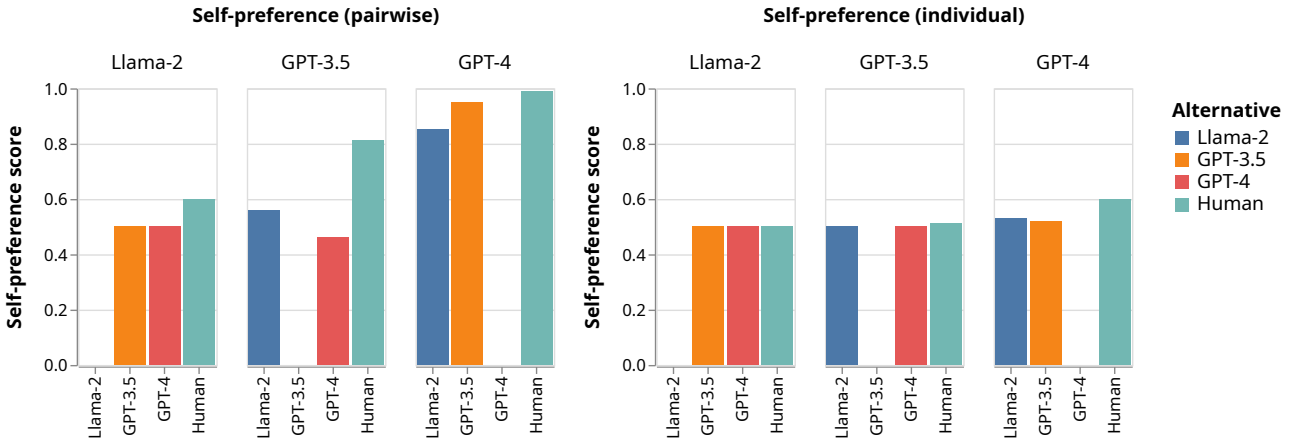

Figure 4. Self-preference before fine-tuning, evaluated using pairwise (left) and individual (right) measurements; scores are aggregated over two datasets.

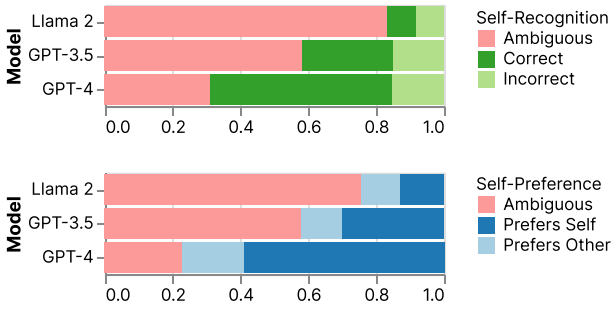

Figure 5. Evaluator responses when treated as a binary response without considering confidence scores. Results are marked "ambiguous" is the response reverses

give very different scores to the different sources, while GPT-4 and GPT-3.5 show signs of self-preference. With the exception of GPT-3.5 on the XSUM dataset, evaluators show a tendency to give the human summary lower scores compared to model-generated summaries and a tendency to give their generations higher scores compared to the scores given by other evaluators (Figure 4.

2.5. Alternative Adjustment for Ordering Bias

All three evaluators models display ordering bias. GPT-4, GPT-3.5, and Llama reverse their pairwise preferences when the ordering of options is reversed at rates of 25%, 58%, and 89% respectively, averaged across tasks and datasets (Figure 5). We account for this bias by averaging the logit-based confidence scores across the two orderings.

An alternative interpretation of the data is, for each evaluator, to discard all the results with the label "ambiguous" where its preference displayed ordering-based reversal, reporting an evaluator's self-recognition ability and self-preference tendency as its frequency of recognizing or preferring its own summary in "unambiguous" cases (Figure 5). This method exposes differences in evaluator results between the two datasets, but supports the presence of out-of-the-box self-recognition and self-preference.

3. Measuring Correlation between Self-Preference and Self-Recognition

Having validated the existence of self-preference and self-recognition, we now turn to study the correlation between the two. The main research question is to understand whether they have a causal relationship; in particular, our hypothesis is that self-recognition causes self-preference—that LLMs prefer their own outputs *because* they recognize them. Our investigation of the hypothesis is motivated by its safety implications, which we discuss in detail in Section 5.

We finetune LLMs to alter their self-recognition capability, and measure how the strength of self-preference changes accordingly. The correlation doesn't by itself prove the causal hypothesis—validating the causal hypothesis would require mechanistic tools that do not exist for LLMs yet (more discussion in Section 5). But we take two approaches to provide more evidence for the hypothesis:

1. We invalidate the inverse causal relationship that self-preference causes self-recognition (Section 3.3).
2. We control for confounders by fine-tuning LLMs on tasks unrelated to self-recognition (Section 3.4).

3.1. Fine-tuning for Self-Recognition

We use supervised fine-tuning on **pairwise examples** to alter the LLMs' self-recognition capability. The training examples are generated in the same way as the evaluation

examples we use in the pairwise measurement. Each example consists of two summaries of a given article, one by the LLM evaluator and the other by another LLM or human source. The evaluators are trained to predict the final token, representing the the correct choice among two options, using supervised learning with cross-entropy loss. To train an LLM evaluator, we use 500 training articles and collect examples from all alternative sources (two other LLMs and human). The remaining 500 articles and the associated summaries are used as the evaluation dataset. We train on the two datasets separately but evaluate both in- and out-of-domain. GPT-3.5 was fine-tuned via API, and Llama 2 using our own implementation. The Llama models are quantized to 8 bits and fine-tuned for one epoch using Adam optimization and a learning rate of 5.0×10^{-5}.

3.2. Fine-Tuning Results

For both detection and comparison tasks in the pairwise setting, ordering bias leads to the average probabilities being the only quantitative distinction in cases where the model favors the second option presented for both orderings of a pairwise question.

Fine-tuning on the self-recognition task causes models to become more confident in their pairwise judgments (Figure 6). When plotting self-recognition score against self-preference, the data clump more tightly around 0.5 and the high- and low-score corners.

In the pairwise setting, fine-tuning for self-recognition results in a strong linear correlation between a model's self-recognition ability and it's self-preference (Figure 7). We find that fine-tuning for self-recognition ability on one dataset transfers to the other.

In additional to analyzing the relationship between self-recognition ability and overall dataset self-preference, we measure the correlation between these two properties on the example level (Table 1). For GPT-3.5 on the XSUM dataset, the evaluator prior to fine-tuning has a correlation of 0.41 (Kendall's τ) between correctly recognizing its summary from a pair and preferring its summary from that same pair. Every fine-tuning configuration we employ results in a model with a positive correlation on this metric and this correlation does not change meaningfully as self-recognition ability increases.

3.3. Invalidating the Inverse Causal Relationship

It's important to ensure that the causal direction is not the reverse—that the LLM does not recognize its own outputs because the quality is objectively higher, in which case the correlation does not reflect favoritism and should not raise any safety concerns. To show that the inverse causal relationship is not the case, we show that LLM evaluators

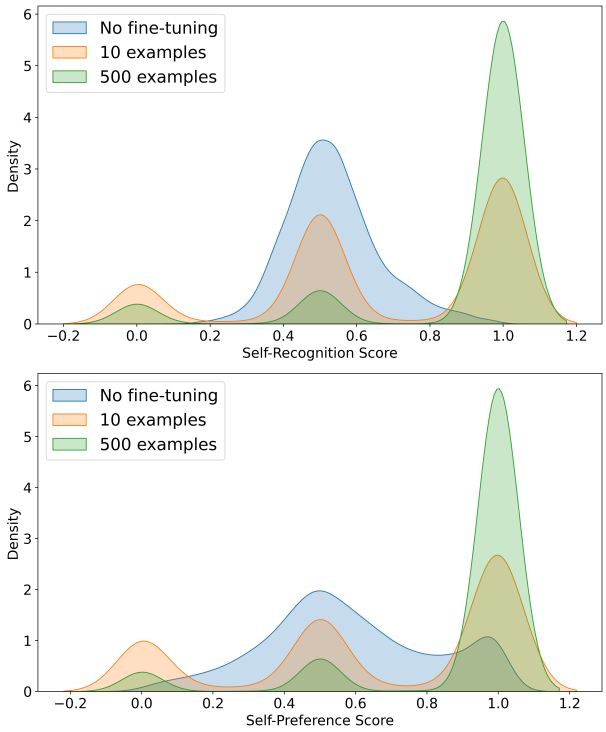

Figure 6. GPT-3.5 pairwise results on XSUM after 0, 10, and 500 fine-tuning examples show a pattern of increasing absolute confidence scores.

do not systematically favor the summaries generated by fine-tuned models to those generated by the original model.

Some fine-tuning runs resulted in degraded generation quality (e.g., summaries that include only the characters "1" and "2"). The remainder show an average preference for the fine-tuned model's generations of 0.46, reflecting a slight preference against the new generations. Only 22% of the evaluator-domain pairs show a preference greater than 0.51 for the fine-tuned model's generations.

3.4. Controlling for Confounders

To confirm that the fine-tuning procedure did not introduce any confounding effect on the variables under study, we additionally fine-tune GPT-3.5 and Llama on three tasks unrelated to self-recognition: length, vowel count, and Flesh-Kincaid readability score (Flesch, 1948). We also fine-tune both models to always output the answer "1" and to answer randomly. Together with the three fine-tuned models trained on the self-recognition task, this gives eight fine-tuned models for each of the two datasets. We find that these models can be placed in the same linear trend that we see in the models fine-tuned on the self-recognition task (Figure 1).

Figure 7. Fine-tuning evaluators for self-recognition results across models and datasets results in a linear trend connecting evaluator self-recognition to self-preference. This effect persists when fine-tuning on one dataset and measuring results on the other dataset.

Model Configuration	XSUM Score	CNN Score
GPT-3.5 Fine-Tuning Runs		
No Fine-Tuning	0.41	0.37
FT (2) on XSUM	0.69	0.64
FT (10) on XSUM	0.66	0.59
FT (500) on XSUM	0.74	0.82
FT (2) on CNN/DM	0.44	0.39
FT (10) on CNN/DM	0.57	0.48
FT (500) on CNN/DM	0.65	0.42
Llama 2 Fine-Tuning Runs		
No Fine-Tuning	0.73	0.50
FT (2) on XSUM	0.60	0.55
FT (10) on XSUM	0.58	0.52
FT (500) on XSUM	0.58	0.51
FT (2) on CNN/DM	0.67	0.54
FT (10) on CNN/DM	0.68	0.61
FT (500) on CNN/DM	0.66	0.59

Table 1. Correlation (Kendall's τ) between the LLM's confidence in recognizing its summary and its confidence in preferring the same summary in pairs of examples.

3.5. Labeling the Source

An additional experiment using only prompting to show evidence of a causal link between self-recognition and self-preference is to add either correct or reversed labels indicating the source of each summary in the pairwise self-preference question (Table 6). The GPT-4 and GPT-3.5 evaluator models show a reversal in self-preference when the labels are reversed in the XSUM dataset; that is, they show a preference for the summary labeled as being their own generation. In the CNN dataset or the Llama evaluator, no clear difference is shown from label reversal.

4. Related Work

4.1. Self-Preference and Biases of Self-Evaluation

The general tendency of LLMs to prefer their own generations was first recognized in the context of LLM-based benchmarks (Bitton et al., 2023a; Zheng et al., 2024; Bai et al., 2024). Liu et al. (2023), similarly to us, study self-preference bias between BERT, T5, and GPT-3.5 on text-summarization. As we discussed The larger capability gap between these models compared to ours make it difficult to control for summarization quality.

Koo et al. (2023) include self-preference in a suite of tests for LLM cognitive biases in a question-answering setting using pairwise measurements. They find GPT-4 to demonstrate lower self-preference than GPT-3.5 out-of-the-box, contrary to our findings, which suggests that evaluation on more datasets is necessary to draw a generalizable conclusion. Neither of these previous works attempted to provide an explanation for self-preference, nor did they study methods to alter self-preference strength.

4.2. Self-recognition and situational awareness

Hoelscher-Obermaier et al. (2023) evaluate GPT-3.5, GPT-4, and Claude-2 on their out-of-the-box self-recognition capabilities. The authors use pairwise measurement on pairs of two ten-sentence fables based on BIG-bench (Srivastava et al., 2023). On this task, contrary to our findings, GPT-3.5 is more accurate than GPT-4, which is less than 50% accurate, again showing the need to experiment on more datasets for generalizable conclusions.

Self-recognition can be seen as a form of situational awareness or self-awareness (Laine et al., 2023; Wang et al., 2024; Berglund et al., 2023; Perez & Long, 2023). Among existing work in this direction, self-recognition is most similar to calibration (Kadavath et al., 2022; Yin et al., 2023; Amayuelas et al., 2023). In this line of work, an LLM is considered to possess self-knowledge if its verbalized uncertainty is well-calibrated. Whereas they measure the correlation between verbalized uncertainty and accuracy, we measure the correlation between the uncertainty on two different tasks.

4.3. LLM detection

Detection of LLM-generated text is important to both AI safety and combating misinformation (Jawahar et al., 2020; Crothers et al., 2023; Wu et al., 2023; Yang et al., 2023;